

St. Thomas' College of Engineering and Technology



PRODUCT RECOMMENDATION SYSTEM USING TF-IDF & COSINE SIMILARITY

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CERTIFICATE

This is to certify that the project report entitled "**Product Recommendation System using TF-IDF & Cosine Similarity**" submitted by Chandni Wakil to the Department of Information Technology in partial fulfillment for the award of the degree of B. Tech in Information Technology is a project work carried out under my supervision during the session 2022–2026. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree.

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Signature with date

Chandni Wakil

PREFACE

This project report presents the work carried out for the development of a **Product Recommendation System Using TF-IDF Vectorization and Cosine Similarity**. The project has been undertaken as a part of the Bachelor of Technology (B.Tech) curriculum in Information Technology to gain practical knowledge of web application development, machine learning, and Natural Language Processing (NLP) techniques.

The primary objective of this project is to develop a simple, efficient, and user-friendly web application that recommends products similar to a selected product based on their textual descriptions. The recommendation engine follows a **content-based filtering** approach by applying **TF-IDF (Term Frequency–Inverse Document Frequency)** to convert product descriptions into numerical feature vectors and **Cosine Similarity** to measure the similarity between products. The system uses a publicly available Kaggle product dataset and generates recommendations without requiring user ratings or purchase history.

The application has been developed using **HTML, CSS, and JavaScript** for the frontend and **Python Flask** for the backend. The recommendation model has been implemented using the **scikit-learn** library, while **pandas** and **NumPy** have been used for data preprocessing and handling. The modular architecture adopted in the project makes the system simple, efficient, and easy to understand for academic purposes.

The development of this project has provided valuable practical experience in integrating web technologies with machine learning techniques. It has strengthened the understanding of content-based recommendation systems, text processing, feature extraction using TF-IDF, similarity measurement using Cosine Similarity, and frontend-backend integration through the Flask framework. The project also demonstrates the practical application of these concepts in addressing real-world product recommendation problems.

VISION & MISSION OF THE INSTITUTE

Vision

To evolve as an industry oriented, research based Institution with ethical values for creative solutions in various engineering domains, with an ultimate objective of meeting technological challenges faced by the Nation and the Society.

Mission

- **M1:** To enhance the quality of engineering education and delivery through accessible, comprehensive and research oriented teaching-learning-assessment processes in the state-of-art environment.
- **M2:** To create opportunities for students and faculty members to acquire professional knowledge with a multidisciplinary approach and develop managerial, entrepreneurial and social attitudes with highly ethical and moral values.
- **M3:** To satisfy the ever-changing needs of the nation with respect to evolution and absorption of sustainable and environment friendly technologies for effective creation of knowledge-based society in the global era.

VISION & MISSION OF THE DEPARTMENT

Vision

To promote the advancement of learning in Information Technology through research-oriented dissemination of knowledge with ethical values which will lead to innovative applications of information in industry and society.

Mission

- **M1:** To incubate the students to grow into industry ready professionals, proficient research scholars and enterprising entrepreneurs.
- **M2:** To create a learner-centric and sustainable environment that motivates the students in adopting emerging technologies of the rapidly changing information society.
- **M3:** To foster innovative responsiveness to social, environmental, and technological challenges among faculty and students.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

Graduates of the Information Technology Program shall:

PEO1: Excel in professional career, higher education, and research with a strong foundation in core and allied areas of Information Technology.

PEO2: Demonstrate professionalism, entrepreneurship, ethical behaviour, effective communication, and collaborative teamwork to adapt to emerging trends through lifelong learning.

PEO3: Exhibit the skills and knowledge required to design, develop, and implement sustainable and ethically sound IT solutions by adopting a multidisciplinary approach to solve real-life problems.

PROGRAM OUTCOMES (PO)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop solutions of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling, recognizing their limitations.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large.

PO10: Project Management and Finance: Apply knowledge of engineering management principles and economic decision-making to one's own work as a member/leader in a team.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for independent and life-long learning and adaptability to new and emerging technologies.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROJECT MAPPING WITH PROGRAM OUTCOMES

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
3	3	2	2	3	1	1	2	2	1	2	2

Correlation levels: 1 – Slight (Low) 2 – Moderate (Medium) 3 – Substantial (High)

Justification:

PO1: Engineering Knowledge (Correlation Level: 3 – Substantial)

The project applies engineering fundamentals, programming concepts, and machine learning techniques to develop a content-based Product Recommendation System. Mathematical concepts such as TF-IDF vectorization and Cosine Similarity are used to convert textual data into numerical feature vectors and measure similarity between products. The implementation using Python, Flask, HTML, CSS, JavaScript, and scikit-learn demonstrates the practical application of Information Technology knowledge to solve a real-world engineering problem. Therefore, the project has a **Substantial (Level 3)** correlation with **PO1**.

PO2: Problem Analysis (Correlation Level: 3 – Substantial)

The project analyses the problem of ineffective product discovery in e-commerce platforms by studying existing recommendation techniques and identifying the limitations of traditional keyword-based search. A detailed literature review was conducted before selecting the most suitable recommendation approach. The project analyses product descriptions, applies TF-IDF feature extraction, and evaluates similarity using Cosine Similarity to generate relevant recommendations. Hence, the project shows a **Substantial (Level 3)** correlation with **PO2**.

PO3: Design and Development of Solutions (Correlation Level: 2 – Moderate)

The project involves designing and developing a complete web-based recommendation system consisting of a user interface, recommendation engine, and result display module. The frontend communicates with the Flask application, which processes recommendation requests and returns suitable products. The developed system provides an effective solution for recommending similar products based on textual descriptions. Therefore, the project has a **Moderate (Level 2)** correlation with **PO3**.

PO4: Investigation of Complex Problems (Correlation Level: 2 – Moderate)

The project investigates different recommendation approaches and compares their suitability before selecting the content-based recommendation technique. Product descriptions are analysed, converted into feature vectors, and evaluated using similarity measures. Testing and result analysis were also performed to verify the effectiveness of the recommendation system. Hence, the project demonstrates a **Moderate (Level 2)** correlation with **PO4**.

PO5: Engineering Tool Usage (Correlation Level: 3 – Substantial)

The project makes effective use of modern engineering and Information Technology tools including Python, Flask, Visual Studio Code, pandas, NumPy, and the scikit-learn library. TF-IDF Vectorizer and Cosine Similarity were implemented using scikit-learn to develop the recommendation engine.

These tools were successfully integrated to build and evaluate the complete system, resulting in a **Substantial (Level 3)** correlation with **PO5**.

PO6: The Engineer and the World (Correlation Level: 1 – Slight)

The proposed recommendation system improves product discovery and helps users find relevant products more efficiently, thereby enhancing the online shopping experience. Although the project has limited direct impact on environmental or societal issues, it contributes towards improving digital services and user convenience. Therefore, the project shows a **Slight (Level 1)** correlation with **PO6**.

PO7: Ethics (Correlation Level: 1 – Slight)

The project uses a publicly available Kaggle dataset and follows ethical practices in data usage. Appropriate references have been provided for all research papers, books, and online resources used during the development of the project. Since the project does not collect or misuse users' personal information, it demonstrates basic professional and ethical responsibility. Hence, the project has a **Slight (Level 1)** correlation with **PO7**.

PO8: Individual and Collaborative Team Work (Correlation Level: 2 – Moderate)

The project was completed through effective collaboration among team members by dividing responsibilities related to literature review, system design, implementation, documentation, testing, and report preparation. Team coordination and communication ensured the successful completion of the project within the specified timeline. Therefore, the project demonstrates a **Moderate (Level 2)** correlation with **PO8**.

PO9: Communication (Correlation Level: 2 – Moderate)

The project involves effective communication through technical documentation, report writing, system diagrams, result presentation, and project demonstration. The developed web interface also provides a simple and user-friendly interaction for displaying recommendations. Therefore, the project shows a **Moderate (Level 2)** correlation with **PO9**.

PO10: Project Management and Finance (Correlation Level: 1 – Slight)

The project required basic planning, task scheduling, and resource management during its development. Open-source software and freely available datasets were used, making the project economical and suitable for academic implementation. Although financial management was not a major component, the project demonstrates basic project planning skills. Hence, the project has a **Slight (Level 1)** correlation with **PO10**.

PO11 : Life-Long Learning (Correlation Level: 2 – Moderate)

The development of this project required learning new technologies such as Flask, TF-IDF Vectorization, Cosine Similarity, and machine learning libraries beyond the regular curriculum. Researching these concepts and implementing them in a practical application encouraged independent learning and technical skill development. Therefore, the project demonstrates a **Moderate (Level 2)** correlation with **PO11**.

PO12 : Life-Long Learning and Adaptability (Correlation Level: 2 – Moderate)

The project encouraged continuous learning by exploring modern recommendation techniques, Natural Language Processing concepts, and machine learning methods applicable to e-commerce systems. The acquired knowledge can be extended to future technologies such as deep learning and hybrid recommendation systems, demonstrating adaptability to technological advancements. Hence, the project has a **Moderate (Level 2)** correlation with **PO12**.

PROGRAM SPECIFIC OUTCOMES (PSO)

On completion of the program the graduates will be able to:

PSO1 (Professional Skill): Apply their knowledge in the field of information technology and contribute to providing appropriate solutions to engineering problems, establishing skills in programming, data-driven computing and machine intelligence.

PSO2 (Competency): Demonstrate proficiency in analytical and critical thinking, methodologies of practical design, and data analysis and interpretation through technical expertise.

PROJECT MAPPING WITH PROGRAM SPECIFIC OUTCOMES

PSO1	PSO2
3	3

Correlation levels: 1 – Slight (Low) 2 – Moderate (Medium) 3 – Substantial (High)

Justification:

PSO1: (Professional Skill) – Correlation Level: 3 (Substantial)

The proposed **Product Recommendation System** strongly satisfies **PSO1** by applying core concepts of Information Technology to develop a practical web-based machine learning application. The project integrates Python, Flask, HTML, CSS, JavaScript, and the scikit-learn library to design and implement a content-based recommendation system. TF-IDF vectorization is used to convert textual product descriptions into numerical feature vectors, while Cosine Similarity is employed to identify products with similar characteristics. The project demonstrates programming skills, data processing, machine learning concepts, and web application development to provide an effective solution for product recommendation. Hence, the project exhibits a **substantial (Level 3)** correlation with PSO1.

PSO2: (Competency) – Correlation Level: 3 (Substantial)

The project strongly supports **PSO2** by demonstrating analytical thinking, data analysis, and practical problem-solving skills. It involves preprocessing product descriptions, extracting meaningful textual features using TF-IDF, computing similarity scores using Cosine Similarity, and analysing the recommendation results to identify the most relevant products. The complete recommendation pipeline requires logical reasoning, algorithm implementation, data interpretation, and performance evaluation through testing and result analysis. The project also includes system design, workflow modelling, testing, and comparative analysis, reflecting the practical application of technical

knowledge to solve a real-world problem. Therefore, the project has a **substantial (Level 3)** correlation with PSO2.

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INTRODUCTION

The rapid growth of e-commerce platforms has made it easier for customers to access a wide variety of products through online shopping. However, the increasing number of available products has also made it difficult for users to identify items that best match their needs. Searching through large product collections can be time-consuming, and traditional keyword-based search methods often fail to provide relevant results when different words are used to describe similar products. As a result, recommendation systems have become an important part of modern e-commerce platforms by helping users discover products that are relevant to their interests (Aggarwal, 2016).

Recommendation systems are commonly classified into **Content-Based Filtering, Collaborative Filtering, and Hybrid Recommendation Systems**. Among these approaches, **Content-Based Filtering** recommends products by analyzing their characteristics or descriptions instead of relying on user ratings or purchase history. This makes it a suitable approach for applications where user interaction data is limited (Pazzani & Billsus, 2007; Lops, de Gemmis, & Semeraro, 2011).

This project presents a **web-based Product Recommendation System** that uses **TF-IDF (Term Frequency–Inverse Document Frequency)** and **Cosine Similarity** to recommend similar products based on their descriptions. **TF-IDF** identifies the important words in product descriptions by assigning greater importance to meaningful terms, while **Cosine Similarity** measures the similarity between product descriptions represented as feature vectors (Salton & Buckley, 1988; Manning, Raghavan, & Schütze, 2008).

The proposed system provides a simple web interface where users can select a product and receive recommendations for similar products. The recommendation process is based on textual information available in product descriptions obtained from a Kaggle dataset. The project demonstrates the practical application of content-based recommendation techniques for improving product discovery in an easy-to-understand and user-friendly manner (Aggarwal, 2016).

This report presents the complete development of the proposed system, including the problem statement, literature review, system design, implementation, testing, result analysis, and future scope. The project demonstrates how **TF-IDF** and **Cosine Similarity** can be effectively applied to build a simple and practical product recommendation system suitable for a B.Tech final-year project (Manning, Raghavan, & Schütze, 2008; Aggarwal, 2016).

CHAPTER 1

PROBLEM STATEMENT, OBJECTIVE & TOOLS

1.1 Problem Statement

The increasing number of products available on e-commerce platforms makes it difficult for users to find relevant products quickly. Traditional keyword-based search methods often return inaccurate results because they rely on exact word matching and may fail to identify products with similar meanings described using different words. This creates the need for a **Product Recommendation System** that can recommend similar products based on their **Product Descriptions**. The proposed system uses **Content-Based Recommendation** by applying **TF-IDF (Term Frequency–Inverse Document Frequency)** to identify important words in product descriptions and **Cosine Similarity** to measure the similarity between products. This approach helps users discover relevant products more efficiently, reduces search effort, and improves the overall shopping experience (Aggarwal, 2016; Manning, Raghavan, & Schütze, 2008).

1.2 Problem Definition

A **Product Recommendation System** is designed to help users discover products that are similar to the one they are interested in, making it easier to find relevant items from a large product collection. Such systems improve product discovery and reduce the time required to search for suitable products (Aggarwal, 2016; Ricci, Rokach, & Shapira, 2015).

TF-IDF (Term Frequency–Inverse Document Frequency) is a text feature extraction technique that determines the importance of words in a document by considering both their frequency within the document and their occurrence across all documents. It helps represent product descriptions as numerical feature vectors for similarity analysis (Salton & Buckley, 1988; Manning, Raghavan, & Schütze, 2008).

Cosine Similarity is a mathematical measure used to determine the similarity between two text vectors. A value closer to **1** indicates higher similarity, while a value closer to **0** indicates lower similarity. It is widely used for comparing textual information in content-based recommendation systems (Manning, Raghavan, & Schütze, 2008).

Based on these concepts, the problem addressed in this project is to recommend products with similar descriptions by applying **Content-Based Recommendation** techniques. Instead of relying on user ratings or purchase history, the system compares **Product Descriptions** using **TF-IDF** and **Cosine Similarity** to identify and recommend relevant products. This approach improves product discovery and helps users find similar products more efficiently (Pazzani & Billsus, 2007; Lops, de Gemmis, & Semeraro, 2011).

1.3 Objective

The primary objective of this project is to develop a **Content-Based Product Recommendation System** that recommends similar products based on their descriptions. The specific objectives of the project are:

- To develop a **Product Recommendation System** using **TF-IDF** and **Cosine Similarity** for recommending similar products.
- To recommend relevant products based on the similarity of product descriptions.
- To improve product discovery by providing meaningful product recommendations.
- To reduce the time and effort required by users to search for similar products.
- To provide a simple and user-friendly web application for displaying recommended products.
- To analyse the recommendations generated by the proposed system and discuss its overall performance and effectiveness.

1.4 Tools and Platform

1.4.1 Frontend

- HTML5, CSS3, JavaScript — for building the user interface
- VS Code — code editor
- Google Chrome — browser for testing

1.4.2 Programming Language

- Python 3.8+

1.4.3 Machine Learning Libraries

- pandas — data loading and preprocessing
- NumPy — numerical operations
- scikit-learn — TfidfVectorizer and cosine_similarity

1.4.4 Web Framework

- Flask — lightweight web framework used to serve the frontend and the recommendation endpoint

1.4.5 Dataset

- Kaggle Product Dataset (CSV containing product titles and descriptions), loaded directly using pandas

1.4.6 Operating System

- Windows 10/11 (all components are cross-platform and also run on Linux/macOS)

1.5 Brief Discussion on Problem

The rapid expansion of e-commerce websites has significantly increased the number of products available to customers. Although this provides users with a wider range of choices, it also makes finding the most relevant products more difficult. Many online shopping platforms still rely on traditional keyword-based search, which mainly matches the exact words entered by the user. As a result, products that are similar in features but described using different words may not appear in the search results, making product discovery less effective.

Another challenge is that product descriptions are written in natural language, where similar products may use different words or phrases. Because of these variations, it is difficult to determine product similarity by directly comparing the text. Therefore, an effective recommendation system requires a method that can identify important information from product descriptions and measure similarity in a meaningful way.

An effective product recommendation system helps users find relevant products more quickly and reduces the time required to search through a large product catalogue. It also improves product discovery by recommending similar products that users might otherwise overlook, thereby enhancing the overall shopping experience (Aggarwal, 2016).

To address this problem, the proposed system applies **TF-IDF (Term Frequency–Inverse Document Frequency)** to assign suitable importance to the words present in product descriptions while reducing the influence of frequently occurring terms (Salton & Buckley, 1988; Manning, Raghavan, & Schütze, 2008). The similarity between products is then measured using **Cosine Similarity**, which compares the textual representations of product descriptions to identify related products (Manning, Raghavan, & Schütze, 2008). This content-based approach enables the system to recommend similar products without depending on user ratings or purchase history, thereby improving product discovery and reducing users' search effort (Aggarwal, 2016; Pazzani & Billsus, 2007).

1.6 Recommendation Systems

A recommendation system is a software application that suggests items to users based on specific criteria or available information. It helps users discover products, services, or content that are relevant to their interests without requiring them to manually search through a large collection of items. Recommendation systems are widely used in e-commerce websites, online learning platforms, music streaming applications, and video streaming services to improve user experience and assist users in making better decisions (Aggarwal, 2016).

Recommendation systems generally work by identifying relationships between items or users and then generating suggestions based on those relationships. Depending on the technique used, recommendations may be generated using product information, user preferences, purchase history, ratings, or browsing behaviour. Recommendation techniques are broadly classified into content-based, collaborative, and hybrid approaches (**Ricci et al., 2015**).

The primary objective of a recommendation system is to reduce the effort required to find relevant products while improving the overall user experience. In this project, recommendations are generated by comparing product descriptions instead of using user purchase history or ratings, making it suitable for content-based recommendation (**Pazzani & Billsus, 2007**).

1.7 Proposed Approach

In a content-based recommendation system, each product is represented using its own information, such as the product title, category, and description. Products that share similar textual characteristics receive higher similarity scores and are recommended to the user. Content-based recommendation is particularly useful when recommendations are generated using product information rather than user interaction data (**Pazzani & Billsus, 2007; Lops et al., 2011**).

The proposed project follows a content-based recommendation approach because the selected Kaggle dataset contains detailed textual descriptions for each product. These descriptions are processed using **TF-IDF (Term Frequency–Inverse Document Frequency)** to generate numerical feature vectors. **Cosine Similarity** is then used to compare these vectors and identify products with similar textual content (**Salton & Buckley, 1988; Manning et al., 2008**).

Since the recommendations are generated using only the information available in the product dataset, the system does not require user ratings, purchase history, or browsing history. This makes the proposed approach simple, efficient, and suitable for content-based product recommendation (**Aggarwal, 2016**).

1.8 Scope of the Project

The scope of this project is limited to the development of a **Content-Based Product Recommendation System** for academic purposes. The application recommends similar products by analysing the textual descriptions available in a Kaggle product dataset.

The system allows users to browse the available products and obtain recommendations based on the selected product. The recommendation engine is implemented using **TF-IDF Vectorization** and **Cosine Similarity**, enabling the system to identify products with similar textual descriptions (**Salton & Buckley, 1988; Manning et al., 2008**).

The application is developed using **Python Flask** for the backend and **HTML, CSS, and JavaScript** for the frontend. It is designed to run on a local system and demonstrates the practical implementation of Natural Language Processing (NLP) techniques within a lightweight web application (**Pallets Projects, 2024**).

The project does not include user authentication, shopping cart functionality, payment processing, order management, or online deployment, as these features are beyond the scope of the present work.

1.9 Applications

- The proposed Product Recommendation System can be applied in the following areas:
- E-commerce platforms for recommending similar products.
- Online shopping websites to improve product discovery.
- Digital product catalogues for searching related products.
- Retail inventory systems to assist customers in finding alternative products.
- Educational projects for demonstrating content-based recommendation techniques.
- Research and learning in the fields of Natural Language Processing (NLP) and Machine Learning (**Aggarwal, 2016; Lops et al., 2011**).

CHAPTER 2

CONCEPTS AND PROBLEM ANALYSIS

2.1 Literature Review

Recommendation systems have become an essential component of modern e-commerce platforms because they help users discover products that match their interests while reducing the time required to search through large product catalogues. These systems improve the overall shopping experience by providing relevant product suggestions based on available information. According to **(Aggarwal, 2016)**, recommendation systems play an important role in information filtering and personalization across various online applications.

Content-Based Recommendation is another widely used approach in which products are recommended based on their own characteristics rather than user behaviour. **(Pazzani and Billsus, 2007)** explain that content-based filtering analyses product features or descriptions to identify similar items. **(Lops, de Gemmis, and Semeraro, 2011)** also state that this approach is effective when user interaction data is limited because recommendations are generated directly from the available product information.

For recommendation systems that use textual information, TF-IDF (Term Frequency–Inverse Document Frequency) is one of the most commonly used feature extraction techniques. It converts product descriptions into numerical feature vectors by assigning higher importance to meaningful words while reducing the influence of frequently occurring words **(Salton & Buckley, 1988; Manning, Raghavan, & Schütze, 2008)**. After feature extraction, Cosine Similarity is widely used to compare these vectors and measure how closely two product descriptions are related **(Manning, Raghavan, & Schütze, 2008)**.

Several research studies have shown that combining TF-IDF with Cosine Similarity provides an efficient method for identifying similar text documents. These techniques are computationally simple, easy to implement, and produce meaningful similarity scores without requiring complex machine learning models or large amounts of training data. This makes them suitable for applications where recommendations are generated using only product descriptions.

Based on the literature reviewed, Content-Based Recommendation using TF-IDF and Cosine Similarity is considered an appropriate approach for this project because the Kaggle dataset contains product descriptions but does not include user ratings or purchase history. Therefore, these techniques

have been adopted to recommend similar products, improve product discovery, and reduce users' search effort (Aggarwal, 2016).

2.2 Comparative Analysis of Recommendation Techniques

Recommendation Technique	Basis	Data Required	Complexity
Content-Based Filtering (TF-IDF)	Product descriptions	Product descriptions	Low
Collaborative Filtering	User ratings and behaviour	User interaction data	Medium
Hybrid Recommendation	Product descriptions and user behaviour	Product descriptions and user interaction data	High

Table 2.1: Comparative Analysis of Recommendation Techniques

Table 2.1 presents a comparison of the major recommendation techniques used in recommendation systems. **Content-Based Filtering** recommends products by analysing their descriptions, whereas **Collaborative Filtering** depends on user ratings and previous interactions. **Hybrid Recommendation** combines both approaches to improve recommendation quality but requires additional data and a more complex implementation (Aggarwal, 2016; Burke, 2002).

Since this project uses only product descriptions from a Kaggle dataset and does not use user ratings or purchase history, **Content-Based Filtering** was selected (Pazzani & Billsus, 2007). **TF-IDF** is used to represent product descriptions as numerical feature vectors, and **Cosine Similarity** is used to identify similar products based on their textual content (Lops, de Gemmis, & Semeraro, 2011).

2.3 Technical Concepts

2.3.1 TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF (Term Frequency–Inverse Document Frequency) is a text feature extraction technique used to determine the importance of words in a document relative to a collection of documents. It assigns higher weights to meaningful words while reducing the importance of commonly occurring words. In this project, TF-IDF is used to convert product descriptions into numerical feature vectors so that the similarity between products can be measured (Salton & Buckley, 1988; Manning, Raghavan, & Schütze, 2008).

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

where:

$$\text{IDF}(t) = \log (N / \text{df}(t))$$

- $\text{TF}(t, d)$ = Frequency of term t in document d
- N = Total number of documents
- $\text{df}(t)$ = Number of documents containing the term t

2.3.2 Cosine Similarity

Cosine Similarity is a mathematical measure used to determine how similar two documents are by comparing their feature vectors. The similarity value ranges from **0 to 1**, where a value closer to **1** indicates higher similarity and a value closer to **0** indicates lower similarity. In this project, Cosine Similarity is used to compare the TF-IDF feature vectors of product descriptions and recommend products with similar textual content (**Manning, Raghavan, & Schütze, 2008**).

The Cosine Similarity formula is:

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \times \|B\|}$$

Where:

- A = Feature vector of the first product
- B = Feature vector of the second product

2.3.3 Content-Based Recommendation

Content-Based Recommendation is a recommendation technique that suggests products by analysing their characteristics or descriptions instead of relying on user ratings or purchase history. Products with similar textual descriptions are considered related and are recommended to the user. Since this project uses only product descriptions from a Kaggle dataset, the content-based recommendation approach is adopted using **TF-IDF** and **Cosine Similarity** to identify similar products (**Pazzani & Billsus, 2007; Lops, de Gemmis, & Semeraro, 2011**).

2.4 Existing System

Existing product recommendation methods used by many e-commerce platforms mainly rely on traditional keyword-based search or collaborative filtering techniques. In keyword-based search, products are retrieved only when the search keywords entered by the user closely match the words present in the product title or description. Although this method is simple and computationally efficient, it often fails to recommend products that have similar features but are described using different words or phrases. As a result, users may not receive the most relevant product suggestions, reducing the effectiveness of product discovery (**Manning, Raghavan, & Schütze, 2008**).

Collaborative Filtering is another widely used recommendation approach in which products are suggested based on user ratings, purchase history, and browsing behaviour. While collaborative filtering can provide personalized recommendations, it requires a large amount of historical user interaction data. The approach also suffers from the cold-start problem, where meaningful recommendations cannot be generated for new users or newly added products because sufficient interaction data is unavailable **(Ricci, Rokach, & Shapira, 2015)**.

Many publicly available recommendation systems also require complex databases, continuous user activity tracking, and regular model updates. Such requirements increase implementation complexity and are not always suitable for academic projects that focus on demonstrating the basic principles of recommendation systems. Therefore, for applications where only product information is available, a content-based recommendation approach provides a simpler and more practical solution **(Pazzani & Billsus, 2007)**.

2.5 Proposed System

The proposed Product Recommendation System follows a content-based recommendation approach in which products are recommended based on the similarity of their textual descriptions. The system uses a publicly available Kaggle product dataset containing product information such as Product ID, Product Title, Category, Price, and Product Description. Among these attributes, the product description is used as the primary feature for generating recommendations.

Initially, the product dataset is loaded into the Flask application and the product descriptions are preprocessed by handling missing values and preparing the text for analysis. TF-IDF (Term Frequency–Inverse Document Frequency) is then applied to convert each product description into a numerical feature vector that represents the importance of individual words. After feature extraction, Cosine Similarity is used to compare the selected product with all other products in the dataset and calculate similarity scores between them **(Salton & Buckley, 1988; Manning, Raghavan, & Schütze, 2008)**.

Whenever a user selects a product through the web interface, the system retrieves the Top-5 products having the highest similarity scores and displays them as recommendations. Since the proposed system depends only on product descriptions, it does not require user ratings, purchase history, or customer profiles. This makes the system suitable for situations where user interaction data is unavailable while still providing meaningful recommendations using textual information **(Aggarwal, 2016)**.

2.6 Advantages of Proposed System

The proposed Product Recommendation System offers the following advantages:

1. **Content-Based Recommendations:**

The system recommends products based on the similarity of product descriptions rather than relying on exact keyword matching, enabling users to discover more relevant products (**Pazzani & Billsus, 2007**).

2. **No User History Required:**

The recommendation process does not depend on user ratings, purchase history, or browsing behaviour. Therefore, the system can generate recommendations even for new users and newly added products (**Aggarwal, 2016**).

3. **Improved Product Discovery:**

By analysing the textual content of product descriptions, the system helps users identify products with similar features, improving the overall product search experience.

4. **Computationally Efficient:**

TF-IDF vectorization and Cosine Similarity are lightweight algorithms that provide fast recommendation results with relatively low computational requirements (**Salton & Buckley, 1988**).

5. **Simple and Easy to Maintain:**

The system is developed using Python, Flask, HTML, CSS, and JavaScript, resulting in a simple architecture that is easy to understand, maintain, and modify.

6. **Suitable for Academic Projects:**

Since the system uses only a publicly available Kaggle dataset without requiring complex databases or user interaction data, it is well suited for the scope of a B.Tech final-year project.

7. **Scalable Design:**

The system can be enhanced in the future by integrating collaborative filtering, user authentication, or advanced Natural Language Processing techniques without changing the overall system architecture.

2.7 Limitations of Existing System

Although the proposed system performs effectively for content-based recommendation, it has certain limitations:

1. **Dependent on Product Descriptions:**

The quality of recommendations depends largely on the completeness and accuracy of product descriptions. Poor or incomplete descriptions may reduce recommendation quality.

2. **No Personalization:**

Since the system does not consider user preferences, ratings, or purchase history, it provides the same recommendations to all users for a selected product.

3. **Limited Semantic Understanding:**

TF-IDF identifies word importance based on frequency but cannot fully understand the contextual meaning of words or complex semantic relationships (Manning, Raghavan, & Schütze, 2008).

4. **Cold Start for Products with Minimal Information:**

Products having very short or insufficient descriptions may not generate highly accurate recommendations because limited textual information is available for comparison.

5. **Limited to Textual Features:**

The system considers only product descriptions and does not utilize additional information such as product images, customer reviews, product ratings, or brand popularity.

6. **Dataset Dependency:**

The recommendation quality depends on the quality and diversity of the Kaggle dataset used. A small or less diverse dataset may reduce the effectiveness of recommendations.

7. **Suitable for Small and Medium Datasets:**

Although efficient for academic purposes, recommendation time may increase when processing very large product datasets unless additional optimization techniques are applied.

CHAPTER 3

DESIGN AND METHODOLOGY

3.1 System Architecture

The proposed system follows a simple web-based architecture for recommending similar products. The user interacts with the system through a web interface developed using **HTML, CSS, and JavaScript**. The recommendation logic is implemented using **Python** and **Flask**, while the product information is obtained from a **Kaggle CSV dataset**. The system uses **TF-IDF** to extract important features from product descriptions and **Cosine Similarity** to compare the similarity between products. Based on the similarity scores, the system recommends products that are closely related to the selected product. Since the product information is stored in a CSV dataset, a separate database is not required for this project.

The overall workflow is straightforward. The user selects a product from the web page, and the selected product information is processed by the recommendation module. The system compares the product description with other product descriptions in the dataset using **TF-IDF** and **Cosine Similarity**. Finally, the most relevant products are displayed on the web page as recommendations, providing users with an easier way to discover similar products.

3.2 Project Workflow

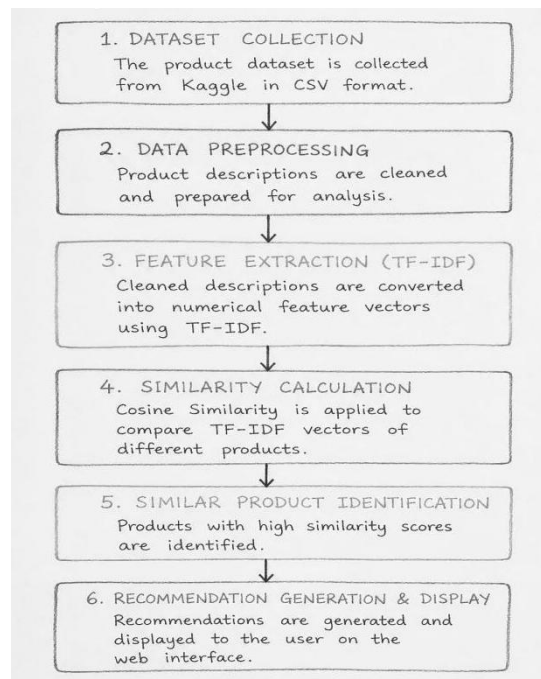


Fig 3.1: Project Workflow

The workflow of the proposed **Product Recommendation System** consists of the following steps:

Step 1: Dataset Collection

The product dataset is collected from **Kaggle** in CSV format. It contains product information such as product names and product descriptions, which are used for generating recommendations.

Step 2: Data Preprocessing

The product descriptions are cleaned and prepared for analysis by removing unnecessary characters and formatting the text into a suitable form for processing.

Step 3: Feature Extraction using TF-IDF

The cleaned product descriptions are converted into numerical feature vectors using **TF-IDF (Term Frequency–Inverse Document Frequency)**. This helps identify the important words in each product description.

Step 4: Similarity Calculation

Cosine Similarity is applied to compare the TF-IDF feature vectors of different products. Based on the similarity scores, products with similar descriptions are identified.

Step 5: Recommendation Generation

When a user selects a product, the system compares it with other products in the dataset and generates a list of similar products based on the calculated similarity scores.

Step 6: Display of Recommendations

The recommended products are displayed through the web interface, allowing users to easily explore products that are similar to the selected item.

3.3 Dataset Description

The proposed Product Recommendation System uses a product dataset collected from Kaggle and stored in CSV format. The dataset contains information such as Product ID, Product Title, Category, Price, and Product Description. The product description field is used for TF-IDF

feature extraction and similarity computation, while the remaining attributes are displayed to the user along with the recommended products.

	A	B	C	D	E
0	Product ID	Product Title	Category	Price (₹)	Product Description
1	101	Wireless Bluetooth Headphones	Electronics	2499	Noise-cancelling wireless Bluetooth headphones with deep bass and 30-hour battery life.
2	102	Bluetooth Earbuds	Electronics	1899	Compact Bluetooth earbuds with charging case, touch controls, and HD stereo sound.
3	103	Gaming Headset	Electronics	2799	Over-ear gaming headset with built-in microphone, RGB lighting, and surround sound.
4	104	Men's Cotton T-Shirt	Fashion	699	Slim-fit cotton T-shirt suitable for casual wear with soft breathable fabric.
5	105	Running Shoes	Footwear	2299	Lightweight sports shoes with breathable mesh upper and anti-slip rubber sole.
6	106	Laptop Backpack	Bags	1499	Water-resistant backpack with padded laptop compartment and USB charging port.
7	107	Stainless Steel Water Bottle	Home & Kitchen	499	Vacuum-insulated stainless steel bottle that keeps beverages hot and cold for hours.
8	108	Wireless Optical Mouse	Electronics	799	Ergonomic wireless optical mouse with adjustable DPI and USB receiver.
9	109	Smart Fitness Band	Electronics	1999	Fitness tracker with heart rate monitoring, sleep tracking, and activity recording.
10	110	Portable Bluetooth Speaker	Electronics	1599	Portable Bluetooth speaker with powerful bass, waterproof design, and long battery backup.

Fig 3.2: Snapshot of the Product Dataset Used in the Recommendation System

The following table shows a sample of the product dataset used in the recommendation system.

3.4 Recommendation Flow Diagram

The Recommendation Flow Diagram illustrates the sequence of operations performed by the system to generate product recommendations. It begins with the user's product selection and follows the processing of the selected product description using **TF-IDF** and **Cosine Similarity** to identify similar products. Finally, the recommended products are displayed on the web interface. This diagram provides a clear understanding of the overall recommendation process used in the proposed system.

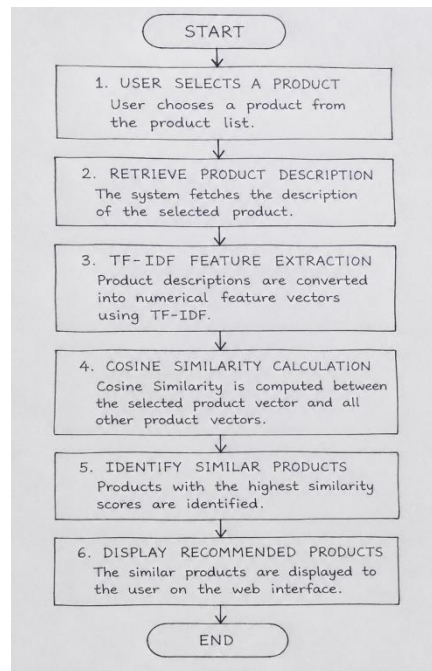


Figure 3.3: Recommendation Flow Diagram

3.5 Data Preprocessing

Before generating product recommendations, the product dataset must be prepared to ensure that the textual information is suitable for feature extraction and similarity computation. Data preprocessing is an important step because raw datasets often contain missing values, inconsistent text formatting, and unnecessary information that may affect the quality of recommendations. In this project, a publicly available Kaggle product dataset was used, containing attributes such as Product ID, Product Title, Category, Price, and Product Description.

The dataset was first loaded into the Flask application using the **pandas** library. The **Product Description** field was selected as the primary attribute for recommendation because it contains the textual information required to identify similarities between products. Missing descriptions were replaced with empty strings using the `fillna("")` function to prevent errors during TF-IDF vectorization.

After handling missing values, the product descriptions were processed using the **TfidfVectorizer** available in the **scikit-learn** library. Common English stop words such as *the*, *is*, *and*, and *of* were automatically removed by specifying the `stop_words='english'` parameter. The vectorizer was configured with a maximum of **5,000 features**, ensuring that only the most informative terms were included in the final feature representation.

The generated TF-IDF matrix was stored in memory when the Flask application started. This preprocessing step was performed only once, allowing subsequent recommendation requests to use the precomputed feature vectors directly. As a result, recommendation generation became faster and avoided repeated computation during every user request (Salton & Buckley, 1988; scikit-learn developers, 2024).

3.6 Dataset Processing Flow

The Dataset Processing Flow illustrates how the product information is processed to generate recommendations. The product dataset is collected from Kaggle in CSV format and is loaded into the system. The product descriptions are then cleaned and preprocessed to remove unnecessary text and prepare them for analysis. Next, **TF-IDF (Term Frequency–Inverse Document Frequency)** is applied to convert the product descriptions into numerical feature vectors. **Cosine Similarity** is then used to measure the similarity between products based on these feature vectors. Finally, the system identifies the most similar products and displays them as recommendations through the web interface. This workflow represents the complete data processing sequence followed by the proposed recommendation system.

3.7 TF-IDF Working with Example

TF-IDF (Term Frequency–Inverse Document Frequency) is a feature extraction technique that converts textual information into numerical feature vectors. Instead of treating every word equally, TF-IDF assigns greater importance to words that occur frequently within a particular document but appear less frequently across the entire dataset. Consequently, meaningful words receive higher weights while common words contribute less to the similarity calculation (Salton & Buckley, 1988).

In this project, each product description is considered as an individual document. After preprocessing, TF-IDF computes a numerical representation of every product description, allowing products to be compared mathematically. The resulting vectors form the basis for calculating product similarity using Cosine Similarity.

The TF-IDF value is calculated using the following relationship:

$$\text{TF-IDF} = \text{Term Frequency (TF)} \times \text{Inverse Document Frequency (IDF)}$$

Where:

- **TF (Term Frequency)** represents the number of times a term appears in a product description.

- **IDF (Inverse Document Frequency)** measures how unique a term is across the complete dataset.

Example

Consider the following two product descriptions:

Product A:

"Wireless Bluetooth Headphones with Noise Cancellation"

Product B:

"Bluetooth Earbuds with Wireless Connectivity"

After removing common stop words, the remaining important terms are:

Product Important Words

Product A wireless, bluetooth, headphones, noise, cancellation

Product B bluetooth, wireless, earbuds, connectivity

Both descriptions contain the words **wireless** and **bluetooth**, which receive meaningful TF-IDF weights because they describe important product features. As a result, both products obtain similar feature vectors and are identified as related products by the recommendation system.

This numerical representation enables the system to compare thousands of product descriptions efficiently while maintaining good recommendation accuracy (Manning, Raghavan, & Schütze, 2008).

3.8 Cosine Similarity Working with Example

After TF-IDF feature vectors have been generated, the similarity between products is measured using **Cosine Similarity**. Instead of comparing product descriptions directly, Cosine Similarity compares the angle between two TF-IDF vectors in a multidimensional feature space. A smaller angle indicates greater similarity between the products (**Manning, Raghavan, & Schütze, 2008**).

The Cosine Similarity value ranges between **0** and **1**.

- **1** indicates that both products are highly similar.
- **0** indicates that the products are completely unrelated.

The mathematical expression is given by:

$$\text{Cosine Similarity} = (\mathbf{A} \cdot \mathbf{B}) / (\|\mathbf{A}\| \times \|\mathbf{B}\|)$$

where **A** and **B** represent the TF-IDF feature vectors of two products.

Example

Suppose the system computes similarity scores for a selected product as follows:

Compared Product Cosine Similarity Score

Bluetooth Earbuds	0.93
Gaming Headset	0.86
Wireless Speaker	0.79
Cotton T-Shirt	0.08

From the above example, the Bluetooth Earbuds have the highest similarity score because their product description shares more meaningful terms with the selected product. In contrast, the Cotton T-Shirt receives a very low similarity score because it belongs to an unrelated category and has completely different textual features.

The recommendation system ranks all similarity scores in descending order and displays the **Top-5** products having the highest scores as recommendations. This approach enables fast and accurate retrieval of similar products without requiring user ratings or purchase history (**Aggarwal, 2016**).

3.9 Module Description

The proposed Product Recommendation System has been developed using a modular architecture in which each module performs a specific function. This modular design improves code readability, simplifies maintenance, and allows future enhancements without affecting the overall functionality of the system.

Module 1: Dataset Loading Module

This module loads the Kaggle product dataset using the **pandas** library. Product information such as Product ID, Product Title, Category, Price, and Product Description is imported into a DataFrame for further processing.

Module 2: Data Preprocessing Module

The preprocessing module prepares the dataset by handling missing values, selecting the Product Description field, and performing TF-IDF vectorization. This module generates the numerical feature vectors required for similarity computation.

Module 3: Recommendation Engine

The recommendation engine forms the core of the proposed system. It computes Cosine Similarity between the selected product and all other products in the dataset using the precomputed TF-IDF matrix. The products with the highest similarity scores are selected as recommendations.

Module 4: Flask Web Application

The Flask application serves as the backend of the system. It loads the dataset, initializes the TF-IDF model during application startup, receives recommendation requests from the frontend, performs similarity computation, and returns the recommendation results in JSON format.

Module 5: Frontend User Interface

The frontend has been developed using **HTML, CSS, and JavaScript**. It displays the product catalogue in a responsive layout and allows users to select products. JavaScript communicates with the Flask backend using HTTP requests and dynamically displays the recommended products without requiring page refresh.

Module 6: Recommendation Display Module

The final module receives the recommendation results returned by the Flask application and presents them as product cards containing the product image, title, category, and price. This module provides a simple and user-friendly interface that enables users to explore similar products efficiently.

3.10 Hardware and Software Requirements

Minimum Hardware

- **Processor:** Intel Core i3 or equivalent
- **RAM:** 4 GB
- **Storage:** 2 GB free disk space

Recommended Hardware

- **Processor:** Intel Core i5 or equivalent
- **RAM:** 8 GB or higher
- **Storage:** 5 GB free disk space

Software Requirements

Component	Software/Technology
Operating System	Windows 10/11 or Ubuntu Linux
Programming Language	Python 3.8 or above
Web Framework	Flask
Frontend	HTML5, CSS3, JavaScript
Machine Learning Libraries	pandas, NumPy, scikit-learn
Development Environment	Visual Studio Code (VS Code)
Dataset	Kaggle Product Dataset (CSV)

CHAPTER 4

CONTROL FLOW & SAMPLE CODES

4.1 Control Flow Diagram

The Control Flow Diagram illustrates the sequence of operations performed by the proposed Product Recommendation System. It shows how the system processes a user's request, verifies the selected product, generates TF-IDF feature vectors, computes cosine similarity, ranks the products based on similarity scores, and finally displays the Top-5 recommended products. It also includes the error handling path when the selected product is not available in the dataset.

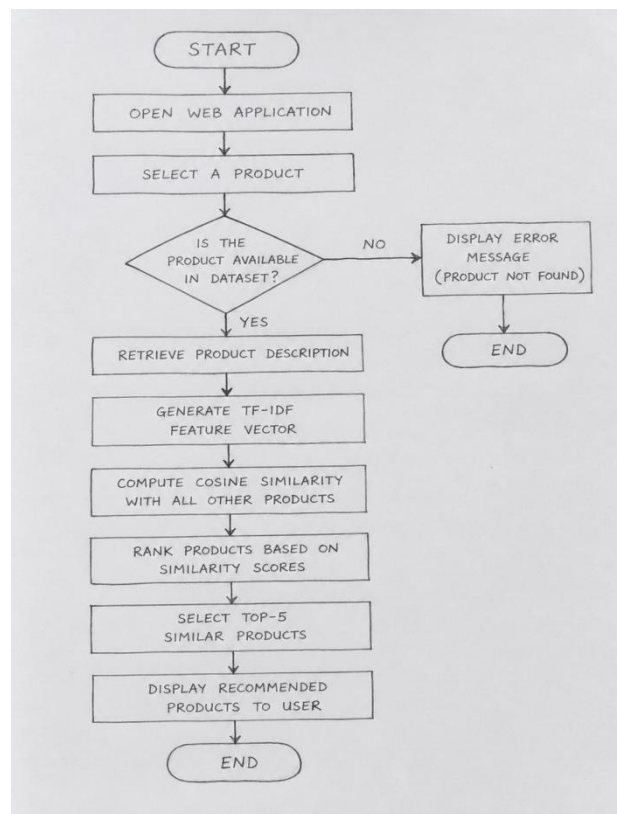


Fig 4.1: Control Flow Diagram of the Product Recommendation System

Explanation of the above diagram: The recommendation process begins when the user opens the web application and selects a product. The system first checks whether the selected product exists in the dataset. If the product is not found, an error message is displayed. Otherwise, the corresponding product description is processed using TF-IDF to generate feature vectors. Cosine Similarity is then calculated between the selected product and all other products. Based on the similarity scores, the

products are ranked, and the Top-5 most similar products are displayed to the user as recommendations.

4.2 Important Code Snippets

- Load dataset `df = pd.read_csv("products.csv")`
- Replace missing descriptions `df['description'] = df['description'].fillna("")`
- Create TF-IDF vectors `vectorizer = TfidfVectorizer(stop_words='english') tfidf_matrix = vectorizer.fit_transform(df['description'])`
- Calculate cosine similarity `similarity_matrix = cosine_similarity(tfidf_matrix)`
- Return Top-5 recommendations `similar_products = similarity_matrix[index].argsort()[::-1][1:6]`

4.2.1 Code Snippet - Calling the Recommendation Engine

```
recommendations = recommend_products(product_id)

return render_template(
    "product.html",
    product=product,
    recommendations=recommendations
)
```

This snippet demonstrates how the Flask backend communicates with the recommendation engine. The generated recommendations are passed to the product page and displayed to the user.

4.4 Code Snippet - TF-IDF Vectorization

```
recommendation.py

vectorizer = TfidfVectorizer()

tfidf_matrix = vectorizer.fit_transform(
    products["description"]
)
```

This snippet converts textual product descriptions into numerical feature vectors that can be compared mathematically.

4.5 Code Snippet - Cosine Similarity

```
similarity = cosine_similarity(
    tfidf_matrix,
```

```
tfidf_matrix  
)
```

This snippet calculates similarity scores between all product vectors, enabling the system to identify products with similar descriptions.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 General

This chapter presents the results obtained after implementing the proposed **Product Recommendation System**. The application was successfully developed and executed on a local system using **Python Flask**, **HTML**, **CSS**, **JavaScript**, and a **Kaggle product dataset**. The recommendation engine was implemented using **TF-IDF (Term Frequency–Inverse Document Frequency)** and **Cosine Similarity** to identify products with similar textual descriptions.

The objective of this chapter is to demonstrate the functionality of the developed application through implementation screenshots and discuss the recommendation results generated by the proposed system. The results confirm that the application successfully recommends similar products based on textual similarity, thereby improving product discovery and reducing the user's search effort.

5.2 Home Page

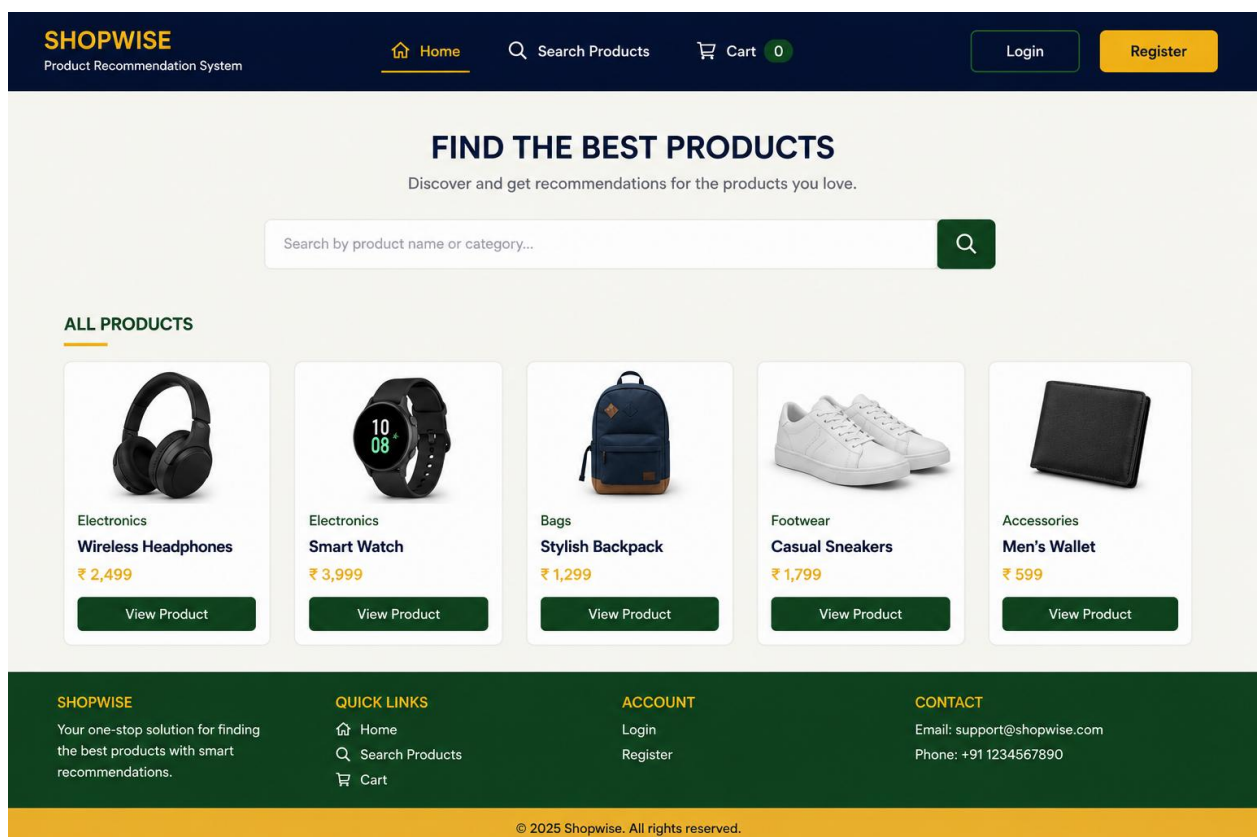


Figure 5.1: Home Page

The Home Page is the first interface displayed when the application is launched. It presents the available products loaded from the Kaggle dataset in a simple and user-friendly layout. Each product is displayed as a separate card, allowing users to browse the available products conveniently.

Each product card displays the following information:

- Product Image
- Product Name
- Product Category
- Product Price
- Product Description

The homepage provides a clean interface that enables users to select any product for viewing its complete details and obtaining similar product recommendations.

5.3 Product Details



Figure 5.2: Product Details

When the user selects a product from the homepage, the Product Details page displays complete information about the selected product. The displayed information helps the user understand the characteristics of the selected product before viewing the recommendations.

The Product Details page displays the following information:

- Product Image
- Product Name
- Product Category
- Product Price
- Product Description

After the product details are displayed, the recommendation engine automatically processes the product description using TF-IDF Vectorization and prepares it for similarity comparison with other products available in the dataset.

5.4 Product Recommendation

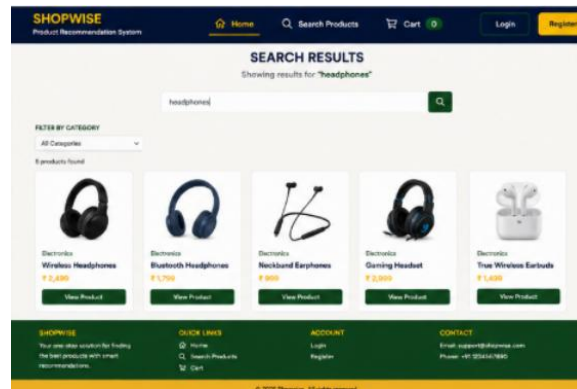


Figure 5.3: Product Recommendation

The recommendation engine generates similar product recommendations using TF-IDF Vectorization and Cosine Similarity. The selected product description is converted into a numerical feature vector, and its similarity is computed with all other products present in the dataset.

Based on the calculated similarity scores, the system displays the Top-5 most similar products. Since the recommendations are generated using only the textual descriptions of products, the proposed system does not require user ratings, purchase history, or browsing history.

The generated recommendations help users discover relevant products efficiently while reducing the time required to search through a large product catalogue.

5.5 Comparative Analysis

Feature	Content-Based (Proposed System)	Recommendation Traditional Search	Keyword
Recommendation Basis	Product descriptions	Exact matching	keyword
User Purchase History Required	No	No	
Ability to Recommend Similar Products	Yes	Limited	
Suitable for New Users	Yes	Yes	
Recommendation Method	TF-IDF and Cosine Similarity	Keyword matching	

Table 5.5: Comparative Analysis of Recommendation Approaches**Comparative Analysis**

The proposed system uses a **Content-Based Recommendation** approach, where products are recommended by analyzing their textual descriptions using **TF-IDF** and **Cosine Similarity**. Unlike traditional keyword-based search, which depends on exact keyword matching, the proposed approach considers the overall textual similarity between product descriptions. As a result, it is able to recommend products with similar features even when different words are used in their descriptions. Since the system does not require user ratings or purchase history, it is also suitable for new users and newly added products. This makes the proposed approach simple, effective, and appropriate for the scope of this academic project (Aggarwal, 2016; Pazzani & Billsus, 2007).

5.6 Discussion on Results

The developed Product Recommendation System successfully recommends products with similar textual descriptions by applying TF-IDF Vectorization and Cosine Similarity. During implementation, the application generated relevant recommendations for different products available in the Kaggle dataset. Products belonging to similar categories and containing similar descriptive keywords consistently appeared among the recommended results.

The recommendation quality mainly depends on the information available in the product descriptions. Products with detailed descriptions produced more meaningful recommendations than products containing limited textual information. Since the proposed system relies entirely on product content, it provides recommendations without requiring any user interaction data, making it suitable for content-based recommendation tasks.

The implementation demonstrates that Natural Language Processing techniques can be effectively applied to develop a simple and efficient recommendation system for academic and small-scale e-commerce applications.

5.7 Limitations of the Proposed System

Although the proposed system successfully recommends similar products, it has certain limitations.

- The recommendation process depends entirely on the textual descriptions available in the dataset.
- Products with very short or incomplete descriptions may produce less accurate recommendations.
- The system recommends products only on the basis of content similarity and does not consider user preferences or purchase history.
- Collaborative filtering and hybrid recommendation techniques have not been implemented.

- The application is designed for local execution and has not been deployed on a cloud platform.

5.8 Performance Analysis

The developed Product Recommendation System performed efficiently during implementation on the selected Kaggle dataset. The TF-IDF Vectorization process successfully converted product descriptions into numerical feature vectors, while Cosine Similarity accurately identified products with similar textual content.

The application generated recommendations within a short time during normal execution and consistently displayed the Top-5 most relevant products for the selected item. The implementation remained stable throughout the execution process and demonstrated that the proposed content-based recommendation approach is suitable for recommending products using only textual information.

The overall performance of the developed system indicates that TF-IDF and Cosine Similarity provide a simple, reliable, and computationally efficient solution for content-based product recommendation in academic applications.

CHAPTER 6

FUTURE SCOPE AND CONCLUSION

6.1 Scope for Future Improvement

The current system recommends products based on their textual descriptions using TF-IDF and Cosine Similarity. In the future, the project can be further improved in the following ways:

- User ratings and purchase history can be incorporated to provide more personalized product recommendations.
- More advanced Natural Language Processing techniques can be explored to improve the accuracy of recommendations.
- A search feature and additional product filters can be added to make product discovery easier for users.
- The recommendation system can be extended to support larger product datasets with improved performance.
- The web interface can be further enhanced to provide a better user experience and a more attractive design.
- The system can be deployed online so that users can access it through the internet.

6.2 Conclusion

This project successfully developed a **Product Recommendation System** using **TF-IDF Vectorization** and **Cosine Similarity** to recommend similar products based on their textual descriptions. The system was implemented using **HTML, CSS, and JavaScript** for the frontend, **Python Flask** for the backend, and the **scikit-learn** library for the recommendation engine. A publicly available Kaggle product dataset was used for generating recommendations.

The developed application successfully converts product descriptions into numerical feature vectors using TF-IDF and measures the similarity between products using Cosine Similarity. Based on these similarity scores, the system recommends the Top-5 most relevant products to the user without requiring user ratings, purchase history, or browsing history.

The generated recommendations were manually verified using sample products from different categories and were found to be relevant in most cases. The developed system demonstrates that content-based recommendation techniques can effectively improve product discovery by recommending products with similar textual characteristics.

The project provided valuable practical experience in web application development, Natural Language Processing, feature extraction, and machine learning. It also strengthened the understanding of integrating frontend and backend technologies with a recommendation engine. Overall, the objectives of the project were successfully achieved, making the proposed system a simple, efficient, and suitable academic implementation of a content-based product recommendation system (**Aggarwal, 2016; Manning, Raghavan, & Schütze, 2008**).

ANNEXURE

This annexure describes the software requirements, dataset details, and execution procedure followed for developing and running the proposed Product Recommendation System.

A. Software Requirements

The following software and libraries were used during the development and execution of the project:

- Python 3.8 or above
- Flask
- pandas
- NumPy
- scikit-learn
- Visual Studio Code (VS Code)
- Modern web browser (Google Chrome or Microsoft Edge)

B. Dataset

The proposed system uses a publicly available **Kaggle Product Dataset** in CSV format. The dataset contains product information such as Product ID, Product Title, Category, Price, Product Description, and Image URL. Product descriptions form the primary input for the recommendation engine because the proposed system follows a content-based recommendation approach. Before generating recommendations, the textual descriptions are preprocessed and converted into numerical feature vectors using TF-IDF Vectorization.

C. Project Execution Procedure

The execution of the application follows the steps given below:

- The product dataset is loaded into the Flask application.
- Missing textual values are handled during data preprocessing.
- Product descriptions are converted into TF-IDF feature vectors.
- Cosine Similarity is computed between the selected product and all other products in the dataset.
- The Top-5 most similar products are identified.
- The generated recommendations are displayed through the web interface.

D. Output

The developed application displays the selected product together with the Top-5 recommended products having similar textual descriptions. The recommendations are presented through a simple and user-friendly web interface, allowing users to discover related products quickly and efficiently.

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